

Reinforcement Learning

Assignment 1

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Contents

1	Assignment 1: Reinforcement Learning	2
1.1	Question 1	2
1.2	Question 2	3
1.3	Question 3	3

1 Assignment 1: Reinforcement Learning

1.1 Question 1

Q1: “Prove that the policy iteration algorithm converges to the optimal policy for a finite MDP with a finite action space for each state.”

First of all, its good to define each key word:

Optimal Policy: An optimal policy is a strategy or decision-making rule that maximizes the expected cumulative reward for an agent in a given environment. An optimal policy is one that leads to the highest possible long-term reward for the agent, considering the dynamics of the environment and the available actions.

Finite MDP: A Markov Decision Process (MDP) is used to model the decision-making problems where an agent interacts with an environment over discrete time steps. The keyword "finite" only means that the MDP consists of a limited set of states and actions with respective transition function that defines the probabilities of moving from one state to another given an action, and a reward function that assigns a numerical value to each state-action pair.

Finite Action Space: A finite action space refers to a situation in which an agent has a limited number of possible actions to choose from in each state of the environment. Meaning that, for every state in the Markov Decision Process (MDP), there is a finite set of actions that the agent can take.

Policy iteration contains a few different steps:

1. **Initialization:** Start with an arbitrary policy π_0 and initialize the value function $V_0(s)$ for all states s .
2. **Policy Evaluation:** For the current policy π_k , compute the value function $V_k(s)$ for all states s using the Bellman equation:

$$V_k(s) = \sum_{a \in A} \pi_k(a|s) \left[R(s, a) + \gamma \sum_{s' \in S} P(s'|s, a) V_k(s') \right]$$

3. **Policy Improvement:** Update the policy π_{k+1} by choosing the action that maximizes the expected value for each state:

$$\pi_{k+1}(s) = \arg \max_{a \in A} \left[R(s, a) + \gamma \sum_{s' \in S} P(s'|s, a) V_k(s') \right]$$

1.2 Question 2

Q2: “Prove that the Bellman optimality equation has a unique solution for a finite MDP with a bounded reward function.”

1.3 Question 3

Q3: “Analyze the challenges of using linear function approximation in value-based reinforcement learning. Discuss the potential issues that may arise and propose methods to mitigate them”

2 Practical Part

Actions and rewards describing the grid-world can be found in the course book at page 60.